

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Animal Husbandry & Veterinary Science

Questions: 19

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q2. (a) What do you mean by feeding experiments ? How can digestibility of maize fodder be determined in bullocks? $3+12=15$ An adult sheep ate 2 kg berseem hay containing 90% dry matter and 12% crude protein and excreted 3 kg faeces containing 30% dry matter and 1.5% crude protein. Calculate the digestibility coefficient of dry matter and crude protein of berseem hay. 15 Write the deficiency symptoms of Vitamin E in poultry. 10

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2. [UNKNOWN] paper-i

Q3. (a) What is grading-up ? Explain the advantages and limitations of grading-up in cattle. 20 (b) Explain, Mendel's law of independent assortment with the help of a checker board. 20

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3. [UNKNOWN] paper-i

Q4. (a) Explain pre-natal and post-natal growth. How is the body weight gain estimated from body measurements ? Enumerate the factors affecting the growth of animals. $10+10+5=25$ (b) Discuss the functions of stomach and liver in ruminants. 15 FSI-P-ANVS 2 ===== PAGE 3 ===== SECTION B

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4. [UNKNOWN] paper-i

Q5. Differentiate between the following: $8 \times 5 = 40$ Megaloblastic anaemia and Pernicious anaemia 8 Steaming-up and Flushing 8 Enzymic digestion and Microbial fermentation 8 Adaptation to cold stress and Adaptation to heat stress 8 Sex influenced and Sex limited characters 8

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5. [UNKNOWN] paper-i

Q5. Differentiate between the following: $8 \times 5 = 40$ Qualitative and Quantitative characters 8 Crude protein and True protein 8 Group feeding and Individual feeding 8 Outcrossing and Top crossing 8 Flushing and Fattening 8

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6. [UNKNOWN] paper-i

Q6. (a) What do you understand by Commercial Dairy Farming? Give a plan to begin a commercial dairy farm comprising of 200 Sahiwal cows. 20 (b) Explain the term 'Natural Disasters'. How can the livestock be saved during natural calamities ? Discuss the feeding and managerial practices under such situations. 20

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7. [UNKNOWN] paper-i

Q6. (a) Enumerate the factors affecting the efficiency of a dairy cow. What is personnel management ? Write the functions of management. $10+5+5=20$ (b) Work out a cropping scheme and determine the land requirement for ensuring the supply of green fodder round the year for a herd of 50 cows and their followers. 20

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8. [UNKNOWN] paper-i

Q7. (a) What do you mean by Carbon-Nitrogen Balance? Give the significance of carbon-nitrogen balance studies in animals. Write the merits and demerits of carbon-nitrogen balance in comparison to comparative slaughter technique and respiratory quotient. 20 (b) Prepare a feeding schedule for a crossbred cow weighing 400 kg and yielding 10 kg milk

daily with 4% fat from green berseem (DCP 2%, TDN 12% and DM 20%), wheat straw (DCP 0%, TDN 40% and DM 90%) and concentrate mixture (DCP 12%, TDN 70% and DM 90%). The maintenance requirement is 250 g DCP and 3.25 kg TDN, whereas requirement for 1 kg milk production is 45 g DCP and 300 g TDN. 20 (28. (a) What is Genotype-Environment Interaction ? Write the practical utility of genotype-environment interaction in animal production. 20 (b) What points will you consider while formulating an economic ration for a lactating buffalo ? How will you prepare an economic ration for finishing pigs? 20 PSI-P-AIVS 3

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9. [UNKNOWN] paper-i

Q7. (a) Discuss the feeding schedule for a buffalo calf from birth to 3 months age. 15 Write the interrelationships between minerals and vitamins. 15 What is calorie : protein ratio ? Write the energy and protein requirements of broilers as per BIS. 10

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10. [UNKNOWN] paper-i

Q8. (a) What is heterosis ? Explain the genetic basis of heterosis. 15 Calculate the heterotic effect in F1 and F2 progenies considering the milk yield in 300 days of Sahiwal and Holstein-Friesian cows as 1600 kg and 5600 kg, respectively, and of their F1 crossbreeds as 3900 kg 10 What do you mean by cross-breeding ? Discuss the various types of cross-breeding. Write the merits and demerits of cross-breeding. 3+6+6=15 FSI-P-AHVS 3

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11. [UNKNOWN] paper-ii

1. Write short notes on the following: 8><5=40 Gametogenesis 8 ,Systemic antifungals 8 Types of placenta in domestic animals 8 Pancreatic secretions and .their functions 8 SPCA and its composition and functions 8 Discuss etiology, pathogenesis, clinical findings, diagnosis, treatment and prevention of abomasal displacement in bovines. 1 5 Classify anti-neoplastic agents with examples. Discuss in detail about their mechanism of action, therapeutic uses and adverse effects. 1 5 Describe in brief four biochemical renal function tests with their clinical significance. 1 0 Discuss in detail epidemiology, clinical signs and prevention and control measures of FMD in domestic animals. 1 5 Discuss following blood protozoan diseases i.e. theileriosis and babesiosis in crossbred cattle with respect to their transmission, diagnosis, treatment and control. 15 Discuss predisposing factors, diagnosis and treatment of Milk Fever in cattle and buffaloes. 10 1 fsi-d-ahvs ===== PAGE 2 ===== Discuss in detail the zoonotic disease brucellosis with respect to its etiology, transmission, diagnosis, prevention and control measures at an organized Animal Farm. • 15 Discuss the occupational zoonotic diseases — chlamydiosis, tuberculosis and avian influenza with respect to mode of transmission and preventive measures for the poultry farmers. 15 Explain veterolegal cases and their investigation at the post-mortem as a Veterinarian. 10 SECTION 'II'

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12. [UNKNOWN] paper-ii

Q2. (a) Furnish in detail the techniques for preparation and processing of animal tissues for histo-pathological examination in a laboratory. 15 Describe in detail the factors that influence blood coagulation and the process of coagulation in animal system. 15 Describe in detail about the foetal components of placenta of porcine with a neat sketch. 10

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13. [UNKNOWN] paper-ii

Q3. (a) A carcass of a one-year-old bull was presented to you for post-mortem examination, with the history of sudden onset of lameness, depression, anorexia, high rise of body temperature, hot and painful swelling on shoulder and gluteal region since a day. The affected muscles on its incision exhibited dark red to black coloured foul smelling, serosanguineous fluid having gas bubbles of varying sizes. The thoracic cavity and pericardial sac revealed blood-strained fluid with little amount of fibrin. Marked oedema along with congestion was observed in the lungs. There were haemorrhages on intestinal mucosa. The impression smear's examination of oedematous fluid showed presence of gram positive, rod-shaped organisms. What is your tentative diagnosis and methods for its confirmative diagnosis? 5

Give control measures and treatment schedule for such cases. 10 During the White Revolution, the Government of India had introduced cross-breeding programme towards self-sufficiency in milk production. List out the diseases that emerged in the cross-bred animals and write in detail about the pathogenesis, diagnosis, treatment and control measures of Anaplasmosis. 15 Explain in detail about the collection, methods of preservation and dispatch of various kinds of samples from a dead animal for laboratory investigation especially for suspected poisoning 10 FSI-D-AHVS 2
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14. [UNKNOWN] paper-ii

Q4. (a) Define and classify Zoonosis. Explain in detail the common means of transmission of pathogens from animals to humans and also list out the higher risk of serious illness in zoonotic diseases. 15 What is epidemiological measure ? Write the applications of epidemiology in disease control. 15 Write in detail about the cellular mechanism of drug action. 10 FSI-D-AHVS 3 ===== PAGE 4 ===== SECTION B

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15. [UNKNOWN] paper-ii

5. Write short notes on the following: 8x5=40 Scope of Rabbit meat production and its utilization in India. 8 Post slaughter physico-chemical changes in meat and factors that influence them. 8 5.(c) Describe the manufacture, procedure and differences between evaporated and condensed milk. 8 Define sanitation., What are different methods of sanitation practiced in meat processing plant? 8 Differences in livestock practices in rural and urban areas. 8 Discuss various methods of pasteurization of milk with their advantages and disadvantages. 15 Describe in detail judging and grading of milk products. 15 Define and compare physico-chemical properties of drum-dried and sprays dried milks. Explain drum drying of milk with the help of flow diagram. 10 Discuss in details general principles of ante mortem and post mortem inspection of food animals. 15 Discuss about utilization of edible and non-edible by-products of slaughter house and their socio-economic implications. 15 Discuss briefly following meat preservation methods and their shelf life : curing, chilling, canning and irradiation. • 10 Discuss in detail classification and functions of various extension teaching methods. 15 Discuss in detail key village, ICDP and operation flood schemes for livestock improvement in India. 15 Discuss problems and constraints in transfer of technology for livestock improvement in India. 10 fsi-d-ahvs 2

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16. [UNKNOWN] paper-ii

Q5. Write short notes on the following: 8x5=40 Sources of contamination in clean milk production and how to overcome the above 8 Concept of value addition in meat 8 Why is there an increase in demand for organic milk ? Write in brief about farm fresh milk and organic milk 8 How stomachs and intestines of a slaughtered ruminant are processed to obtain better price ? 8 Constraints in the course of technology dissemination at the farmers' doors 8

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17. [UNKNOWN] paper-ii

Q6. (a) What are the factors deteriorating the quality of market milk? Describe in detail how Codex Alimentarius Commission is helpful in quality control of milk and milk products. 15 Describe in detail about the various processing steps involved in commercial ghee, ice-cream and khoa production and the legal quality standards adopted in India. 15 List out the common adulterants in Indian milk and write about the various methods of detection of milk adulteration at home level. 10

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18. [UNKNOWN] paper-ii

Q7. (a) How can the quality of mutton and chicken be improved with the existing slaughter and meat sale facilities in our country? 15 What are meat products ? Classify meat products and write in detail about the need for processing meat products and their preservation. 15 Sixty-five percent of a buffalo is by-product. Explain, in detail, how an entrepreneur is going to manage a slaughter house profitably. 10 FSI-D-ATIVS 4 ===== PAGE 5 =====

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19. [UNKNOWN] paper-ii

Q8. (a) To improve the socio-economic condition and sustainable livelihood of small-holding farmers, suggest innovative Animal husbandry practices which fit in their resources and environment. 15 How can electronic media in rural conditions serve as an effective source of advice and assistance to farmers to improve their production and marketing of their produce? 15 Write in detail about the merits and demerits of rabbit rearing in India. 10 FSI-D-AHVS 5

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